

```
hadoop@BDTT: ~/kafka_2.13-3.5.1
hadoop@BDTT:~/kafka_2.13-3.5.1$ bin/kafka-server-start.sh config/server.properties
[2026-03-11 18:50:50,935] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2026-03-11 18:50:51,569] INFO Setting -D jdk.tls.rejectClientInitiatedRenegotiation=true to disable client-initiated TLS renegotiat
ion (org.apache.zookeeper.common.X509Util)
[2026-03-11 18:50:51,717] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2026-03-11 18:50:51,719] INFO starting (kafka.server.KafkaServer)
[2026-03-11 18:50:51,720] INFO Connecting to zookeeper on localhost:2181 (kafka.server.KafkaServer)
[2026-03-11 18:50:51,824] INFO [ZooKeeperClient Kafka server] Initializing a new session to localhost:2181. (kafka.zookeeper.ZooKeep
erClient)
[2026-03-11 18:50:51,831] INFO Client environment:zookeeper.version=3.6.4--d65253dc6f8e9097c6e95a126463fd5fdeb4521c, built on 12/18/
2022 18:10 GMT (org.apache.zookeeper.ZooKeeper)
[2026-03-11 18:50:51,831] INFO Client environment:java.version=11.0.20, java.vendor=Oracle Corporation, java.home=/usr/lib/jvm/java-11-openjdk-amd64, os.name=Linux, os.arch=amd64, os.version=5.15.0-92-generic, user.name=hadoop, user.home=/home/hadoop, user.dir=/home/hadoop, java.class.version=11.0.20
```

Creating a Kafka Topic

```
hadoop@BDTT: ~/kafka_2.13-3.5.1
hadoop@BDTT:~/kafka_2.13-3.5.1$ bin/kafka-topics.sh --create --topic lab-file-stream --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic lab-file-stream.
hadoop@BDTT:~/kafka_2.13-3.5.1$ bin/kafka-topics.sh --list --bootstrap-server localhost:9092
lab-file-stream
hadoop@BDTT:~/kafka_2.13-3.5.1$
```

Creating Input and Output Files

```
lab-file-stream
hadoop@BDTT:~/kafka_2.13-3.5.1$ touch input.txt
hadoop@BDTT:~/kafka_2.13-3.5.1$ touch output.txt
hadoop@BDTT:~/kafka_2.13-3.5.1$ ls
bin  config  input.txt  libs  LICENSE  licenses  logs  NOTICE  output.txt  site-docs
hadoop@BDTT:~/kafka_2.13-3.5.1$
```

Configuring the File Source Connector

```
hadoop@BDTT:~/kafka_2.13-3.5.1$ nano config/connect-file-source.properties
hadoop@BDTT:~/kafka_2.13-3.5.1$
```

Defining File Source Connector Properties

```
hadoop@BDTT: ~/kafka_2.13-3.5.1
GNU nano 6.2 config/connect-file-source.properties *
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements. See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

name=local-file-source
connector.class=FileStreamSource
tasks.max=1
file=input.txt
topic=lab-file-stream
```

Configuring the File Sink Connector

```
hadoop@BDTT:~/kafka_2.13-3.5.1$ nano config/connect-file-sink.properties
hadoop@BDTT:~/kafka_2.13-3.5.1$ █
```

Defining File Sink Connector Properties

```
hadoop@BDTT: ~/kafka_2.13-3.5.1
GNU nano 6.2 config/connect-file-sink.properties
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements. See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

name=local-file-sink
connector.class=FileStreamSink
tasks.max=1
file=output.txt
topics=lab-file-stream
```

Starting Kafka Connect Worker

```
hadoop@BDTT:~/kafka_2.13-3.5.1$ bin/connect-standalone.sh config/connect-standalone.properties config/connect-file-source.properties
config/connect-file-sink.properties
[2026-03-11 19:26:29,763] INFO Kafka Connect worker initializing ... (org.apache.kafka.connect.cli.AbstractConnectCli:114)
[2026-03-11 19:26:29,774] INFO WorkerInfo values:
  jvm.args = -Xms256M, -Xmx2G, -XX:+UseG1GC, -XX:MaxGCPauseMillis=20, -XX:InitiatingHeapOccupancyPercent=35, -XX:+ExplicitGCIn
vokesConcurrent, -XX:MaxInlineLevel=15, -Djava.awt.headless=true, -Dcom.sun.management.jmxremote, -Dcom.sun.management.jmxremote.aut
henticate=false, -Dcom.sun.management.jmxremote.ssl=false, -Dkafka.logs.dir=/home/hadoop/kafka_2.13-3.5.1/bin/../logs, -Dlog4j.confi
guration=file:bin/../config/connect-log4j.properties
  jvm.spec = Ubuntu, OpenJDK 64-Bit Server VM, 11.0.22, 11.0.22+7-post-Ubuntu-0ubuntu222.04.1
  jvm.classpath = /home/hadoop/kafka_2.13-3.5.1/bin/../libs/activation-1.1.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/aop
alliance-repackaged-2.6.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/argparse4j-0.7.0.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../li
bs/audience-annotations-0.13.0.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/commons-cli-1.4.jar:/home/hadoop/kafka_2.13-3.5.1/bin/.
../libs/commons-lang3-3.8.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/connect-api-3.5.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/.
../libs/connect-basic-auth-extension-3.5.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/connect-json-3.5.1.jar:/home/hadoop/kafka_2.13
-3.5.1/bin/../libs/connect-mirror-3.5.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/connect-mirror-client-3.5.1.jar:/home/hadoop/k
afka_2.13-3.5.1/bin/../libs/connect-runtime-3.5.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/connect-transforms-3.5.1.jar:/home/h
adoop/kafka_2.13-3.5.1/bin/../libs/hk2-api-2.6.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/hk2-locator-2.6.1.jar:/home/hadoop/ka
fka_2.13-3.5.1/bin/../libs/hk2-utils-2.6.1.jar:/home/hadoop/kafka_2.13-3.5.1/bin/../libs/jackson-annotations-2.13.5.jar:/home/hadoop
```

Appending Messages to the Input File

```
hadoop@BDTT: ~/kafka_2.13-3.5.1
hadoop@BDTT:~/kafka_2.13-3.5.1$ echo "First Kafka message" >> input.txt
hadoop@BDTT:~/kafka_2.13-3.5.1$ echo "Second Kafka message" >> input.txt
hadoop@BDTT:~/kafka_2.13-3.5.1$ echo "Third Kafka message" >> input.txt
hadoop@BDTT:~/kafka_2.13-3.5.1$
```

Verification of Kafka Processes and Configuration

```
hadoop@BDTT: /usr/local/kafka/config
hadoop... x hadoop... x hadoop... x hadoop... x hadoop... x v
hadoop@BDTT:/usr/local/kafka$ jps
5650 Kafka
5270 QuorumPeerMain
6071 Jps
hadoop@BDTT:/usr/local/kafka$ bin/kafka-topics.sh --create --topic lab-file-stream --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
Created topic lab-file-stream.
hadoop@BDTT:/usr/local/kafka$ bin/kafka-topics.sh --list --bootstrap-server localhost:9092
lab-file-stream
hadoop@BDTT:/usr/local/kafka$ cd ~
hadoop@BDTT:~$ touch input.txt
hadoop@BDTT:~$ nano input.txt
hadoop@BDTT:~$ touch output.txt
hadoop@BDTT:~$ cd /usr/local/kafka/config
hadoop@BDTT:/usr/local/kafka/config$ nano connect-file-source.properties
hadoop@BDTT:/usr/local/kafka/config$ nano connect-file-sink.properties
hadoop@BDTT:/usr/local/kafka/config$
```

Running Kafka Connect in Standalone Mode

```
hadoop@BDTT: /usr/local/kafka
hadoop@BDTT:/usr/local/kafka/config$ bin/connect-standalone.sh config/connect-standalone.properties config/connect-file-source.properties config/connect-file-sink.properties
bash: bin/connect-standalone.sh: No such file or directory
hadoop@BDTT:/usr/local/kafka/config$ cd /usr/local/kafka
hadoop@BDTT:/usr/local/kafka$ bin/connect-standalone.sh config/connect-standalone.properties config/connect-file-source.properties config/connect-file-sink.properties
[2026-03-12 10:36:00,170] INFO Kafka Connect worker initializing ... (org.apache.kafka.connect.cli.AbstractConnectCli:114)
[2026-03-12 10:36:00,174] INFO WorkerInfo values:
    jvm.args = -Xms256M, -Xmx2G, -XX:+UseG1GC, -XX:MaxGCPauseMillis=20, -XX:InitiatingHeapOccupancyPercent=35, -XX:+ExplicitGCInvokesConcurrent, -XX:MaxInLineLevel=15, -Djava.awt.headless=true, -Dcom.sun.management.jmxremote, -Dcom.sun.management.jmxremote.authenticate=false, -Dcom.sun.management.jmxremote.ssl=false, -Dkafka.logs.dir=/usr/local/kafka/bin/./logs, -Dlog4j.configuration=file:bin/./config/connect-log4j.properties
    jvm.spec = Ubuntu, OpenJDK 64-Bit Server VM, 11.0.22, 11.0.22+7-post-Ubuntu-0ubuntu222.04.1
    jvm.classpath = /usr/local/kafka/bin/./libs/activation-1.1.1.jar:/usr/local/kafka/bin/./libs/aopalliance-repackaged-2.6.1.jar:/usr/local/kafka/bin/./libs/argparse4j-0.7.0.jar:/usr/local/kafka/bin/./libs/audience-annotations-0.13.0.jar:/usr/local/kafka/bin/./libs/commons-cli-1.4.jar:/usr/local/kafka/bin/./libs/commons-lang3-3.8.1.jar:/usr/local/kafka/bin/./libs/connect-api-3.5.1.jar:
```

Verifying the Output and Kafka Topic Messages

```
hadoop@BDTT: ~
hadoop@BDTT:/usr/local/kafka$ cd ~
hadoop@BDTT:~$ echo "Line 1 Kafka Lab" >> input.txt
hadoop@BDTT:~$ echo "Line 2 Kafka Lab" >> input.txt
hadoop@BDTT:~$ echo "Line 3 Kafka Lab" >> input.txt
hadoop@BDTT:~$ cat output.txt
Kafka Lab Line 1
Kafka Lab Line 2
Kafka Lab Line 3
Kafka Lab Line 4
Line 1 Kafka Lab
Line 2 Kafka Lab
Line 3 Kafka Lab
hadoop@BDTT:~$ cd /usr/local/kafka
hadoop@BDTT:/usr/local/kafka$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic lab-file-stream --from-beginning
{"schema":{"type":"string","optional":false},"payload":"Kafka Lab Line 1"}
{"schema":{"type":"string","optional":false},"payload":"Kafka Lab Line 2"}
{"schema":{"type":"string","optional":false},"payload":"Kafka Lab Line 3"}
{"schema":{"type":"string","optional":false},"payload":"Kafka Lab Line 4"}
{"schema":{"type":"string","optional":false},"payload":"Line 1 Kafka Lab"}
{"schema":{"type":"string","optional":false},"payload":"Line 2 Kafka Lab"}
{"schema":{"type":"string","optional":false},"payload":"Line 3 Kafka Lab"}
```